

# Continuous Nuclear Mass-Sector Register from $A = 1$ to $A = 119$

Exploratory Post-Silicon Extension of the CPTG Universal Nuclear-Reaction Program

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## Abstract

This computational addendum documents an exploratory extension of the CPTG nuclear-reaction program beyond the native reaction-network inventory at silicon-30. A prescribed-trajectory graph containing forward neutron-capture and beta-minus edges produces an archived, gap-free mass-sector register from  $A = 1$  through  $A = 119$ . An independent replay reproduces the accepted  $A \leq 120$ , eight-substep result to a maximum relative difference of  $1.75 \times 10^{-15}$ . Moving the absorbing boundary to  $A = 140$  and  $A = 160$  changes every mass-sector sum through  $A = 119$  by exactly zero. Temporal refinement through 64 substeps preserves positive support while changing selected  $A = 31$ – $119$  checkpoints by at most approximately 1.06% from 32 to 64 substeps. Exact zero-source and source-scaling controls establish that the continuation is driven only by the stored native silicon-30 trajectory. Removing its early values below  $10^{-50}$  preserves positive graph support in the enlarged qualification graph but strongly suppresses deep-tail magnitudes. The published register is therefore a robust reduced-graph reachability result through  $A = 119$ , not a precision-qualified prediction of primordial heavy-element abundances or a finite physical endpoint. Because the seed calculation inherits parameters introduced previously in the published CPTG gravitational comparison framework, the extension also provides supporting evidence for cross-scale structural continuity; the full theoretical interpretation remains with the companion universal-theory paper.

## 1 Introduction

This document is a computational addendum to the Curvature Polarization Transport Gravity (CPTG) universal nuclear-reaction program. The program begins with the free-nucleon vertex, proceeds through the deuterium bridge and the mass-three closure pair, and reaches helium-four saturation under the source, amplitude, rate, transport, and abundance-closure structure developed for the companion universal-theory paper. That broader theory originated in the commissioned CPTG nuclear-reaction study [1]. The present addendum follows the represented native reaction-network inventory through oxygen-16 and silicon-30, then documents a prescribed-trajectory neutron-capture and beta-minus continuation of the nuclide graph.

The detailed unpublished coefficient construction of the universal four-sector law is reserved for the companion theory paper and is not repeated here. For masses above  $A = 30$ , this addendum begins from the stored native thermodynamic, neutron, and silicon-30 trajectories and applies selected JINA ReacLib neutron-capture and beta-minus edge rates through `pynucastro` [4, 5]. The post-silicon calculation is an output-layer reachability and numerical-qualification study seeded by the native CPTG result.

The purpose of this extension is to identify, organize, and document the complete set of mass sectors and nuclide states represented in the combined calculation. Every integer mass number in the interval

$$1 \leq A \leq 119 \tag{1}$$

appears exactly once in Table 2. Mass five is included explicitly through the unbound helium-five and lithium-five resonance states relevant to the represented  $n + {}^4\text{He}$  and  $p + {}^4\text{He}$  channels, although neither state is retained as an abundance-bearing network coordinate [6].

The calculation contains three distinct levels of support. Masses  $A = 1\text{--}30$  are included in the native reaction-network inventory. Masses  $A = 31\text{--}32$  are obtained from the numerically refined prescribed-trajectory continuation beyond silicon-30. Masses  $A = 33\text{--}119$  belong to the larger exploratory neutron-capture and beta-minus graph. These categories preserve the distinction between native-network coverage, the strong first post-silicon diagnostic, and the deeper exploratory continuation.

The mass-120 sector in the archived calculation is an absorbing computational boundary and is excluded from physical interpretation. The new qualification moves that boundary outward to  $A = 140$  and  $A = 160$ . The purpose is to determine which parts of the register are stable properties of the selected reduced graph and which depend on temporal resolution or the smallest values in the native silicon-30 source trajectory.

The scientific purpose of this addendum is narrower than that of the companion theory paper. It records the post-silicon computational extension, its mass-sector register, and the controls that determine which parts of the continuation are numerically stable. The broader theoretical interpretation is summarized only to identify the origin of the structure being extended.

## 2 Relation to the CPTG Framework

The comparison parameters carried into the universal nuclear-reaction theory were introduced in the published *CPTG Geometric Pi Branch: Comparison Coordinates* paper [2]. Curvature polarization and curvature transport provide the broader CPTG interpretation of structural response and directed propagation. In the nuclear construction, that distinction is reflected in differentiated reaction-source states and baryon-conserving transport through the ordered light-nuclide sequence.

The resulting gravitational-to-nuclear connection is important because the nuclear program extends previously introduced CPTG structure rather than replacing it with an unrelated sector-specific fit. The present paper does not attempt to establish the complete theoretical priority or interpretation of that transfer. Its task is to document the exploratory continuation seeded by the native silicon-30 result; the companion universal-theory paper remains the authority for the full nuclear law and its cross-scale significance.

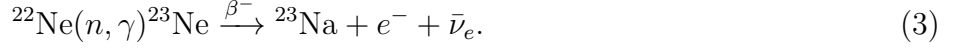
## 3 Evidence Chain and Prescribed-Trajectory Continuation

The universal light-state sequence is

$$\{n, p\} \longrightarrow {}^2\text{H} \longrightarrow \{{}^3\text{H}, {}^3\text{He}\} \longrightarrow {}^4\text{He}. \tag{2}$$

The absence of a persistent mass-five state does not terminate the reaction network. The represented chain bypasses the mass-five resonances and reaches mass six and higher through multi-body and

capture channels. The native basis proceeds through oxygen-16, while the extended native inventory reaches silicon-30 after inclusion of the bridge



The first post-silicon continuation is



The native trajectory was generated with AlterBBN v2.2 at  $\eta_{10} = 6.122$ , with the AlterBBN `failsafe` control set to 6 and the species-abundance floor set to  $10^{-50}$  [3]. Under this configuration, inclusion of the bridge gives

$$Y_{^{23}\text{Na}} = 1.0229842 \times 10^{-31}, \quad (6)$$

$$Y_{^{30}\text{Si}} = 1.0741850 \times 10^{-46}. \quad (7)$$

The compact high-resolution post-silicon diagnostic gives

$$Y_{A=31} = 1.2642940 \times 10^{-47}, \quad (8)$$

$$Y_{A=32} = 7.1605139 \times 10^{-48}. \quad (9)$$

These two values come from the compact  $A \leq 40$ , 32-substep convergence calculation. To keep the continuous register on one numerical basis, Table 2 instead reports the single  $A \leq 120$ , eight-substep values for every diagnostic mass from  $A = 31$  through  $A = 119$ . The corresponding table values at  $A = 31$  and  $A = 32$  are  $1.2608074 \times 10^{-47}$  and  $7.1659590 \times 10^{-48}$ , differing from the compact values by approximately 0.276% and 0.076%, respectively.

The archived prescribed-trajectory calculation was executed with Python 3.13.5 and `pynucastro` 2.11.0. The  $A \leq 120$  graph contains 1,079 reachable species including  $^{30}\text{Si}$  and 2,009 directed reaction edges. The graph retains forward  $(n, \gamma)$  captures and  $\beta^-$  decays only and is integrated by background-interpolated recursive backward Euler on the resulting acyclic graph. The continuous register uses eight integration substeps per native trajectory interval; the compact  $A \leq 40$  convergence calculation uses 32 substeps per interval. Reverse photodisintegration and post-silicon charged-particle channels are not included in this prescribed-trajectory graph.

Every downstream post-silicon species is initialized to exactly zero. The continuation reads only the stored temperature, baryon-density, free-neutron, and  $^{30}\text{Si}$  trajectories. Thus no abundance-floor entries from other native nuclides are imported into the continuation state. A zero- $^{30}\text{Si}$  source control leaves every downstream mass exactly zero, while half-strength and double-strength source controls reproduce exactly one half and twice the baseline output at every mass.

### 3.1 Extended qualification tests

The accepted calculation was extended without changing its physical edge selection or recurrence. The qualification matrix contains:

- a direct replay of the accepted  $A \leq 120$ , eight-substep result;

- outward absorbing boundaries at  $A = 140$  and  $A = 160$ ;
- temporal refinements of 16, 32, and 64 substeps on the  $A \leq 160$  graph;
- exact zero-, half-, and double-source controls;
- source-cutoff controls that set stored  $^{30}\text{Si}$  values below  $10^{-50}$ ,  $10^{-49}$ , or  $10^{-48}$  to zero before integration.

The enlarged  $A \leq 160$  graph contains 1,767 reachable species including  $^{30}\text{Si}$  and 3,316 directed edges. The accepted  $A \leq 120$  result is reproduced with a maximum relative difference of  $1.7494 \times 10^{-15}$ . For every mass sector from  $A = 31$  through  $A = 119$ , the  $A_{\text{max}} = 120, 140$ , and 160 calculations are exactly identical at eight substeps: all three pairwise maximum relative differences are zero, with 89 of 89 mass sectors exactly equal in each comparison.

| $A$ | $Y_A$ (S8)                 | $Y_A$ (S64)                | S32–S64 change | S8–S64 change |
|-----|----------------------------|----------------------------|----------------|---------------|
| 31  | $1.260807 \times 10^{-47}$ | $1.264869 \times 10^{-47}$ | 0.0455%        | 0.3211%       |
| 32  | $7.165959 \times 10^{-48}$ | $7.159464 \times 10^{-48}$ | 0.0147%        | 0.0906%       |
| 40  | $1.687562 \times 10^{-52}$ | $1.567476 \times 10^{-52}$ | 1.0590%        | 7.1159%       |
| 100 | $4.660660 \times 10^{-57}$ | $4.435122 \times 10^{-57}$ | 0.7056%        | 4.8392%       |
| 119 | $2.873667 \times 10^{-57}$ | $2.740476 \times 10^{-57}$ | 0.6751%        | 4.6349%       |

Table 1: Selected temporal-convergence results on the enlarged  $A_{\text{max}} = 160$  graph. S8, S32, and S64 denote integration substeps per stored native trajectory interval.

The maximum selected S32-to-S64 change is approximately 1.06%, at  $A = 40$ . Positive support remains present at every interior mass from  $A = 31$  through  $A = 159$  in the 64-substep enlarged calculation.

The source-cutoff controls sharpen the interpretation. Removing  $^{30}\text{Si}$  source values below  $10^{-50}$  changes  $A = 31$  by only  $6.10 \times 10^{-9}$  relative and  $A = 32$  by approximately  $1.08 \times 10^{-4}$  at eight substeps, but changes  $A = 40$  by approximately 36.8% and suppresses selected sectors from  $A = 56$  upward by more than 99.99%. At 64 substeps with the same cutoff, positive support still reaches every interior mass through  $A = 159$ , while the  $A = 119$  magnitude falls to approximately  $6.42 \times 10^{-75}$ .

Accordingly, the outward-boundary and temporal results strengthen the graph-reachability claim, but the deep-tail magnitudes remain sensitive to the earliest and smallest values in the native  $^{30}\text{Si}$  trajectory. The calculation rules out silicon-30 as a topological endpoint of this selected reduced graph. It does not establish a finite physical endpoint, a precision-qualified heavy-element yield, or native-network coverage beyond silicon-30.

## 4 Continuous Nuclide Register from Mass 1 Through Mass 119

For each abundance-bearing nuclide  $i$ ,  $Y_i$  denotes its endpoint number abundance per baryon. The abundance assigned to mass sector  $A$  is

$$Y_A = \sum_{i:A_i=A} Y_i. \quad (10)$$

Table 2 contains 119 consecutive mass rows and 1,111 listed nuclide or resonance entries. At fixed mass number, entries belonging to different elements are isobars rather than isotopes; the table therefore uses the more precise term *nuclide*, consistent with IUPAC terminology [7]. Masses 1 through 30 reproduce the complete controlling native endpoint inventory. The  $^8\text{Be}$  value in that source is the declared  $10^{-50}$  abundance-floor entry and should not be interpreted as a persistent primordial abundance. Masses 31

through 119 reproduce every node in the one-way prescribed-trajectory neutron-capture and beta-minus graph. A listed diagnostic node records rate-library graph membership; it does not, by itself, establish experimental observation or measurable primordial production. Three listed graph nodes— $^{43}\text{Si}$ ,  $^{47}\text{Si}$ , and  $^{51}\text{Si}$ —have exactly zero endpoint abundance in the eight-substep output; every mass-sector sum from  $A = 31$  through  $A = 119$  remains positive.

The abundance column uses the native endpoint sums for  $A = 1$ –30 and preserves the archived single  $A \leq 120$ , eight-substep diagnostic output for  $A = 31$ –119. Those archived values are retained so that the register remains an exact reproduction object for the original release. Table 1 gives the stronger temporal qualification without silently replacing the archived basis. For D1 and D2 rows,  $Y_A$  is a diagnostic endpoint population in the reduced one-way graph and is not asserted to be a physical primordial heavy-element abundance. The column is marked not applicable at  $A = 5$  because the listed entries are resonances rather than abundance coordinates. Numerical values are reproduced at the precision stored in the controlling source files; variation in printed trailing digits does not represent an independently assigned uncertainty. The evidence-class codes are defined immediately before the register.

**Class codes:** **N0**, native universal light-state core; **N1**, native oxygen-16 basis; **N2**, native post-oxygen extension through silicon-30; **D1**, numerically refined prescribed-trajectory continuation; **D2**, exploratory prescribed-trajectory graph; **R**, listed resonance entries without an abundance coordinate.

Table 2: Gap-free register of the nuclides and resonances represented in the combined calculations for every mass number from  $A = 1$  through  $A = 119$ . The column  $N$  gives the number of listed entries at each mass.

| $A$ | $N$ | Represented nuclides                                | Archived $Y_A$             | Class |
|-----|-----|-----------------------------------------------------|----------------------------|-------|
| 1   | 2   | $n$ , $^1\text{H}(p)$                               | $7.522677 \times 10^{-1}$  | N0    |
| 2   | 1   | $^2\text{H}(\text{D})$                              | $1.860436 \times 10^{-5}$  | N0    |
| 3   | 2   | $^3\text{H}(\text{T})$ , $^3\text{He}$              | $7.789007 \times 10^{-6}$  | N0    |
| 4   | 1   | $^4\text{He}$                                       | $6.191793 \times 10^{-2}$  | N0    |
| 5   | 2   | $^5\text{He}$ , $^5\text{Li}$                       | not applicable             | R     |
| 6   | 1   | $^6\text{Li}$                                       | $8.214136 \times 10^{-15}$ | N1    |
| 7   | 2   | $^7\text{Li}$ , $^7\text{Be}$                       | $4.024159 \times 10^{-10}$ | N1    |
| 8   | 3   | $^8\text{Li}$ , $^8\text{Be}$ , $^8\text{B}$        | $2.246811 \times 10^{-34}$ | N1    |
| 9   | 1   | $^9\text{Be}$                                       | $1.86585 \times 10^{-19}$  | N1    |
| 10  | 2   | $^{10}\text{Be}$ , $^{10}\text{B}$                  | $2.468639 \times 10^{-21}$ | N1    |
| 11  | 2   | $^{11}\text{B}$ , $^{11}\text{C}$                   | $3.495781 \times 10^{-16}$ | N1    |
| 12  | 3   | $^{12}\text{B}$ , $^{12}\text{C}$ , $^{12}\text{N}$ | $3.582536 \times 10^{-16}$ | N1    |

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| $A$ | $N$ | Represented nuclides                                                                       | Archived $Y_A$             | Class |
|-----|-----|--------------------------------------------------------------------------------------------|----------------------------|-------|
| 13  | 2   | $^{13}\text{C}$ , $^{13}\text{N}$                                                          | $7.450479 \times 10^{-17}$ | N1    |
| 14  | 3   | $^{14}\text{C}$ , $^{14}\text{N}$ , $^{14}\text{O}$                                        | $4.090225 \times 10^{-17}$ | N1    |
| 15  | 2   | $^{15}\text{N}$ , $^{15}\text{O}$                                                          | $9.790313 \times 10^{-21}$ | N1    |
| 16  | 1   | $^{16}\text{O}$                                                                            | $3.57075 \times 10^{-20}$  | N1    |
| 17  | 1   | $^{17}\text{O}$                                                                            | $3.134831 \times 10^{-22}$ | N2    |
| 18  | 2   | $^{18}\text{O}$ , $^{18}\text{F}$                                                          | $1.020704 \times 10^{-24}$ | N2    |
| 19  | 1   | $^{19}\text{F}$                                                                            | $2.790054 \times 10^{-27}$ | N2    |
| 20  | 1   | $^{20}\text{Ne}$                                                                           | $1.298913 \times 10^{-28}$ | N2    |
| 21  | 1   | $^{21}\text{Ne}$                                                                           | $5.006889 \times 10^{-30}$ | N2    |
| 22  | 2   | $^{22}\text{Ne}$ , $^{22}\text{Na}$                                                        | $6.868353 \times 10^{-30}$ | N2    |
| 23  | 2   | $^{23}\text{Ne}$ , $^{23}\text{Na}$                                                        | $1.022984 \times 10^{-31}$ | N2    |
| 24  | 1   | $^{24}\text{Mg}$                                                                           | $2.630659 \times 10^{-35}$ | N2    |
| 25  | 1   | $^{25}\text{Mg}$                                                                           | $1.678778 \times 10^{-36}$ | N2    |
| 26  | 2   | $^{26}\text{Mg}$ , $^{26}\text{Al}$                                                        | $2.364879 \times 10^{-36}$ | N2    |
| 27  | 1   | $^{27}\text{Al}$                                                                           | $5.198167 \times 10^{-39}$ | N2    |
| 28  | 1   | $^{28}\text{Si}$                                                                           | $1.14832 \times 10^{-44}$  | N2    |
| 29  | 1   | $^{29}\text{Si}$                                                                           | $3.693064 \times 10^{-46}$ | N2    |
| 30  | 1   | $^{30}\text{Si}$                                                                           | $1.074185 \times 10^{-46}$ | N2    |
| 31  | 2   | $^{31}\text{Si}$ , $^{31}\text{P}$                                                         | $1.260807 \times 10^{-47}$ | D1    |
| 32  | 3   | $^{32}\text{Si}$ , $^{32}\text{P}$ , $^{32}\text{S}$                                       | $7.165959 \times 10^{-48}$ | D1    |
| 33  | 3   | $^{33}\text{Si}$ , $^{33}\text{P}$ , $^{33}\text{S}$                                       | $6.418241 \times 10^{-49}$ | D2    |
| 34  | 3   | $^{34}\text{Si}$ , $^{34}\text{P}$ , $^{34}\text{S}$                                       | $2.919824 \times 10^{-49}$ | D2    |
| 35  | 4   | $^{35}\text{Si}$ , $^{35}\text{P}$ , $^{35}\text{S}$ , $^{35}\text{Cl}$                    | $1.435181 \times 10^{-49}$ | D2    |
| 36  | 5   | $^{36}\text{Si}$ , $^{36}\text{P}$ , $^{36}\text{S}$ , $^{36}\text{Cl}$ , $^{36}\text{Ar}$ | $5.546139 \times 10^{-50}$ | D2    |

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| A  | N  | Represented nuclides                                                                                                                                                                                                            | Archived $Y_A$             | Class |
|----|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------|
| 37 | 5  | $^{37}\text{Si}$ , $^{37}\text{P}$ , $^{37}\text{S}$ , $^{37}\text{Cl}$ , $^{37}\text{Ar}$                                                                                                                                      | $5.235818 \times 10^{-51}$ | D2    |
| 38 | 5  | $^{38}\text{Si}$ , $^{38}\text{P}$ , $^{38}\text{S}$ , $^{38}\text{Cl}$ , $^{38}\text{Ar}$                                                                                                                                      | $2.698372 \times 10^{-51}$ | D2    |
| 39 | 6  | $^{39}\text{Si}$ , $^{39}\text{P}$ , $^{39}\text{S}$ , $^{39}\text{Cl}$ , $^{39}\text{Ar}$ , $^{39}\text{K}$                                                                                                                    | $4.871949 \times 10^{-52}$ | D2    |
| 40 | 7  | $^{40}\text{Si}$ , $^{40}\text{P}$ , $^{40}\text{S}$ , $^{40}\text{Cl}$ , $^{40}\text{Ar}$ , $^{40}\text{K}$ , $^{40}\text{Ca}$                                                                                                 | $1.687562 \times 10^{-52}$ | D2    |
| 41 | 7  | $^{41}\text{Si}$ , $^{41}\text{P}$ , $^{41}\text{S}$ , $^{41}\text{Cl}$ , $^{41}\text{Ar}$ , $^{41}\text{K}$ , $^{41}\text{Ca}$                                                                                                 | $2.217256 \times 10^{-52}$ | D2    |
| 42 | 7  | $^{42}\text{Si}$ , $^{42}\text{P}$ , $^{42}\text{S}$ , $^{42}\text{Cl}$ , $^{42}\text{Ar}$ , $^{42}\text{K}$ , $^{42}\text{Ca}$                                                                                                 | $1.8769 \times 10^{-52}$   | D2    |
| 43 | 7  | $^{43}\text{Si}$ , $^{43}\text{P}$ , $^{43}\text{S}$ , $^{43}\text{Cl}$ , $^{43}\text{Ar}$ , $^{43}\text{K}$ , $^{43}\text{Ca}$                                                                                                 | $8.938921 \times 10^{-53}$ | D2    |
| 44 | 7  | $^{44}\text{Si}$ , $^{44}\text{P}$ , $^{44}\text{S}$ , $^{44}\text{Cl}$ , $^{44}\text{Ar}$ , $^{44}\text{K}$ , $^{44}\text{Ca}$                                                                                                 | $1.271115 \times 10^{-52}$ | D2    |
| 45 | 8  | $^{45}\text{Si}$ , $^{45}\text{P}$ , $^{45}\text{S}$ , $^{45}\text{Cl}$ , $^{45}\text{Ar}$ , $^{45}\text{K}$ , $^{45}\text{Ca}$ , $^{45}\text{Sc}$                                                                              | $5.279289 \times 10^{-53}$ | D2    |
| 46 | 9  | $^{46}\text{Si}$ , $^{46}\text{P}$ , $^{46}\text{S}$ , $^{46}\text{Cl}$ , $^{46}\text{Ar}$ , $^{46}\text{K}$ , $^{46}\text{Ca}$ , $^{46}\text{Sc}$ , $^{46}\text{Ti}$                                                           | $4.728164 \times 10^{-53}$ | D2    |
| 47 | 9  | $^{47}\text{Si}$ , $^{47}\text{P}$ , $^{47}\text{S}$ , $^{47}\text{Cl}$ , $^{47}\text{Ar}$ , $^{47}\text{K}$ , $^{47}\text{Ca}$ , $^{47}\text{Sc}$ , $^{47}\text{Ti}$                                                           | $3.905347 \times 10^{-53}$ | D2    |
| 48 | 9  | $^{48}\text{Si}$ , $^{48}\text{P}$ , $^{48}\text{S}$ , $^{48}\text{Cl}$ , $^{48}\text{Ar}$ , $^{48}\text{K}$ , $^{48}\text{Ca}$ , $^{48}\text{Sc}$ , $^{48}\text{Ti}$                                                           | $3.351341 \times 10^{-53}$ | D2    |
| 49 | 9  | $^{49}\text{Si}$ , $^{49}\text{P}$ , $^{49}\text{S}$ , $^{49}\text{Cl}$ , $^{49}\text{Ar}$ , $^{49}\text{K}$ , $^{49}\text{Ca}$ , $^{49}\text{Sc}$ , $^{49}\text{Ti}$                                                           | $2.322291 \times 10^{-53}$ | D2    |
| 50 | 9  | $^{50}\text{Si}$ , $^{50}\text{P}$ , $^{50}\text{S}$ , $^{50}\text{Cl}$ , $^{50}\text{Ar}$ , $^{50}\text{K}$ , $^{50}\text{Ca}$ , $^{50}\text{Sc}$ , $^{50}\text{Ti}$                                                           | $1.119949 \times 10^{-53}$ | D2    |
| 51 | 10 | $^{51}\text{Si}$ , $^{51}\text{P}$ , $^{51}\text{S}$ , $^{51}\text{Cl}$ , $^{51}\text{Ar}$ , $^{51}\text{K}$ , $^{51}\text{Ca}$ , $^{51}\text{Sc}$ , $^{51}\text{Ti}$ , $^{51}\text{V}$                                         | $6.471204 \times 10^{-54}$ | D2    |
| 52 | 9  | $^{52}\text{S}$ , $^{52}\text{Cl}$ , $^{52}\text{Ar}$ , $^{52}\text{K}$ , $^{52}\text{Ca}$ , $^{52}\text{Sc}$ , $^{52}\text{Ti}$ , $^{52}\text{V}$ , $^{52}\text{Cr}$                                                           | $4.588623 \times 10^{-54}$ | D2    |
| 53 | 9  | $^{53}\text{S}$ , $^{53}\text{Cl}$ , $^{53}\text{Ar}$ , $^{53}\text{K}$ , $^{53}\text{Ca}$ , $^{53}\text{Sc}$ , $^{53}\text{Ti}$ , $^{53}\text{V}$ , $^{53}\text{Cr}$                                                           | $3.067894 \times 10^{-54}$ | D2    |
| 54 | 9  | $^{54}\text{S}$ , $^{54}\text{Cl}$ , $^{54}\text{Ar}$ , $^{54}\text{K}$ , $^{54}\text{Ca}$ , $^{54}\text{Sc}$ , $^{54}\text{Ti}$ , $^{54}\text{V}$ , $^{54}\text{Cr}$                                                           | $1.037553 \times 10^{-54}$ | D2    |
| 55 | 10 | $^{55}\text{S}$ , $^{55}\text{Cl}$ , $^{55}\text{Ar}$ , $^{55}\text{K}$ , $^{55}\text{Ca}$ , $^{55}\text{Sc}$ , $^{55}\text{Ti}$ , $^{55}\text{V}$ , $^{55}\text{Cr}$ , $^{55}\text{Mn}$                                        | $1.169812 \times 10^{-54}$ | D2    |
| 56 | 11 | $^{56}\text{S}$ , $^{56}\text{Cl}$ , $^{56}\text{Ar}$ , $^{56}\text{K}$ , $^{56}\text{Ca}$ , $^{56}\text{Sc}$ , $^{56}\text{Ti}$ , $^{56}\text{V}$ , $^{56}\text{Cr}$ , $^{56}\text{Mn}$ , $^{56}\text{Fe}$                     | $1.765245 \times 10^{-54}$ | D2    |
| 57 | 11 | $^{57}\text{S}$ , $^{57}\text{Cl}$ , $^{57}\text{Ar}$ , $^{57}\text{K}$ , $^{57}\text{Ca}$ , $^{57}\text{Sc}$ , $^{57}\text{Ti}$ , $^{57}\text{V}$ , $^{57}\text{Cr}$ , $^{57}\text{Mn}$ , $^{57}\text{Fe}$                     | $7.957791 \times 10^{-55}$ | D2    |
| 58 | 11 | $^{58}\text{S}$ , $^{58}\text{Cl}$ , $^{58}\text{Ar}$ , $^{58}\text{K}$ , $^{58}\text{Ca}$ , $^{58}\text{Sc}$ , $^{58}\text{Ti}$ , $^{58}\text{V}$ , $^{58}\text{Cr}$ , $^{58}\text{Mn}$ , $^{58}\text{Fe}$                     | $5.671194 \times 10^{-55}$ | D2    |
| 59 | 11 | $^{59}\text{Cl}$ , $^{59}\text{Ar}$ , $^{59}\text{K}$ , $^{59}\text{Ca}$ , $^{59}\text{Sc}$ , $^{59}\text{Ti}$ , $^{59}\text{V}$ , $^{59}\text{Cr}$ , $^{59}\text{Mn}$ , $^{59}\text{Fe}$ , $^{59}\text{Co}$                    | $4.114257 \times 10^{-55}$ | D2    |
| 60 | 12 | $^{60}\text{Cl}$ , $^{60}\text{Ar}$ , $^{60}\text{K}$ , $^{60}\text{Ca}$ , $^{60}\text{Sc}$ , $^{60}\text{Ti}$ , $^{60}\text{V}$ , $^{60}\text{Cr}$ , $^{60}\text{Mn}$ , $^{60}\text{Fe}$ , $^{60}\text{Co}$ , $^{60}\text{Ni}$ | $9.994849 \times 10^{-55}$ | D2    |

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| A  | N  | Represented nuclides                                                                                                                                                                                                                                                   | Archived $Y_A$             | Class |
|----|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------|
| 61 | 12 | $^{61}\text{Cl}$ , $^{61}\text{Ar}$ , $^{61}\text{K}$ , $^{61}\text{Ca}$ , $^{61}\text{Sc}$ , $^{61}\text{Ti}$ , $^{61}\text{V}$ , $^{61}\text{Cr}$ , $^{61}\text{Mn}$ , $^{61}\text{Fe}$ , $^{61}\text{Co}$ , $^{61}\text{Ni}$                                        | $4.205458 \times 10^{-55}$ | D2    |
| 62 | 11 | $^{62}\text{Ar}$ , $^{62}\text{K}$ , $^{62}\text{Ca}$ , $^{62}\text{Sc}$ , $^{62}\text{Ti}$ , $^{62}\text{V}$ , $^{62}\text{Cr}$ , $^{62}\text{Mn}$ , $^{62}\text{Fe}$ , $^{62}\text{Co}$ , $^{62}\text{Ni}$                                                           | $6.831138 \times 10^{-55}$ | D2    |
| 63 | 12 | $^{63}\text{Ar}$ , $^{63}\text{K}$ , $^{63}\text{Ca}$ , $^{63}\text{Sc}$ , $^{63}\text{Ti}$ , $^{63}\text{V}$ , $^{63}\text{Cr}$ , $^{63}\text{Mn}$ , $^{63}\text{Fe}$ , $^{63}\text{Co}$ , $^{63}\text{Ni}$ , $^{63}\text{Cu}$                                        | $3.540761 \times 10^{-55}$ | D2    |
| 64 | 13 | $^{64}\text{Ar}$ , $^{64}\text{K}$ , $^{64}\text{Ca}$ , $^{64}\text{Sc}$ , $^{64}\text{Ti}$ , $^{64}\text{V}$ , $^{64}\text{Cr}$ , $^{64}\text{Mn}$ , $^{64}\text{Fe}$ , $^{64}\text{Co}$ , $^{64}\text{Ni}$ , $^{64}\text{Cu}$ , $^{64}\text{Zn}$                     | $3.755918 \times 10^{-55}$ | D2    |
| 65 | 12 | $^{65}\text{K}$ , $^{65}\text{Ca}$ , $^{65}\text{Sc}$ , $^{65}\text{Ti}$ , $^{65}\text{V}$ , $^{65}\text{Cr}$ , $^{65}\text{Mn}$ , $^{65}\text{Fe}$ , $^{65}\text{Co}$ , $^{65}\text{Ni}$ , $^{65}\text{Cu}$ , $^{65}\text{Zn}$                                        | $1.504781 \times 10^{-55}$ | D2    |
| 66 | 12 | $^{66}\text{K}$ , $^{66}\text{Ca}$ , $^{66}\text{Sc}$ , $^{66}\text{Ti}$ , $^{66}\text{V}$ , $^{66}\text{Cr}$ , $^{66}\text{Mn}$ , $^{66}\text{Fe}$ , $^{66}\text{Co}$ , $^{66}\text{Ni}$ , $^{66}\text{Cu}$ , $^{66}\text{Zn}$                                        | $4.948077 \times 10^{-55}$ | D2    |
| 67 | 12 | $^{67}\text{K}$ , $^{67}\text{Ca}$ , $^{67}\text{Sc}$ , $^{67}\text{Ti}$ , $^{67}\text{V}$ , $^{67}\text{Cr}$ , $^{67}\text{Mn}$ , $^{67}\text{Fe}$ , $^{67}\text{Co}$ , $^{67}\text{Ni}$ , $^{67}\text{Cu}$ , $^{67}\text{Zn}$                                        | $1.517976 \times 10^{-55}$ | D2    |
| 68 | 11 | $^{68}\text{Ca}$ , $^{68}\text{Sc}$ , $^{68}\text{Ti}$ , $^{68}\text{V}$ , $^{68}\text{Cr}$ , $^{68}\text{Mn}$ , $^{68}\text{Fe}$ , $^{68}\text{Co}$ , $^{68}\text{Ni}$ , $^{68}\text{Cu}$ , $^{68}\text{Zn}$                                                          | $3.464737 \times 10^{-55}$ | D2    |
| 69 | 12 | $^{69}\text{Ca}$ , $^{69}\text{Sc}$ , $^{69}\text{Ti}$ , $^{69}\text{V}$ , $^{69}\text{Cr}$ , $^{69}\text{Mn}$ , $^{69}\text{Fe}$ , $^{69}\text{Co}$ , $^{69}\text{Ni}$ , $^{69}\text{Cu}$ , $^{69}\text{Zn}$ , $^{69}\text{Ga}$                                       | $2.851959 \times 10^{-55}$ | D2    |
| 70 | 13 | $^{70}\text{Ca}$ , $^{70}\text{Sc}$ , $^{70}\text{Ti}$ , $^{70}\text{V}$ , $^{70}\text{Cr}$ , $^{70}\text{Mn}$ , $^{70}\text{Fe}$ , $^{70}\text{Co}$ , $^{70}\text{Ni}$ , $^{70}\text{Cu}$ , $^{70}\text{Zn}$ , $^{70}\text{Ga}$ , $^{70}\text{Ge}$                    | $1.086518 \times 10^{-55}$ | D2    |
| 71 | 13 | $^{71}\text{Ca}$ , $^{71}\text{Sc}$ , $^{71}\text{Ti}$ , $^{71}\text{V}$ , $^{71}\text{Cr}$ , $^{71}\text{Mn}$ , $^{71}\text{Fe}$ , $^{71}\text{Co}$ , $^{71}\text{Ni}$ , $^{71}\text{Cu}$ , $^{71}\text{Zn}$ , $^{71}\text{Ga}$ , $^{71}\text{Ge}$                    | $1.460832 \times 10^{-55}$ | D2    |
| 72 | 12 | $^{72}\text{Sc}$ , $^{72}\text{Ti}$ , $^{72}\text{V}$ , $^{72}\text{Cr}$ , $^{72}\text{Mn}$ , $^{72}\text{Fe}$ , $^{72}\text{Co}$ , $^{72}\text{Ni}$ , $^{72}\text{Cu}$ , $^{72}\text{Zn}$ , $^{72}\text{Ga}$ , $^{72}\text{Ge}$                                       | $1.863515 \times 10^{-55}$ | D2    |
| 73 | 12 | $^{73}\text{Sc}$ , $^{73}\text{Ti}$ , $^{73}\text{V}$ , $^{73}\text{Cr}$ , $^{73}\text{Mn}$ , $^{73}\text{Fe}$ , $^{73}\text{Co}$ , $^{73}\text{Ni}$ , $^{73}\text{Cu}$ , $^{73}\text{Zn}$ , $^{73}\text{Ga}$ , $^{73}\text{Ge}$                                       | $8.537773 \times 10^{-56}$ | D2    |
| 74 | 12 | $^{74}\text{Sc}$ , $^{74}\text{Ti}$ , $^{74}\text{V}$ , $^{74}\text{Cr}$ , $^{74}\text{Mn}$ , $^{74}\text{Fe}$ , $^{74}\text{Co}$ , $^{74}\text{Ni}$ , $^{74}\text{Cu}$ , $^{74}\text{Zn}$ , $^{74}\text{Ga}$ , $^{74}\text{Ge}$                                       | $2.819861 \times 10^{-55}$ | D2    |
| 75 | 12 | $^{75}\text{Ti}$ , $^{75}\text{V}$ , $^{75}\text{Cr}$ , $^{75}\text{Mn}$ , $^{75}\text{Fe}$ , $^{75}\text{Co}$ , $^{75}\text{Ni}$ , $^{75}\text{Cu}$ , $^{75}\text{Zn}$ , $^{75}\text{Ga}$ , $^{75}\text{Ge}$ , $^{75}\text{As}$                                       | $1.851494 \times 10^{-55}$ | D2    |
| 76 | 13 | $^{76}\text{Ti}$ , $^{76}\text{V}$ , $^{76}\text{Cr}$ , $^{76}\text{Mn}$ , $^{76}\text{Fe}$ , $^{76}\text{Co}$ , $^{76}\text{Ni}$ , $^{76}\text{Cu}$ , $^{76}\text{Zn}$ , $^{76}\text{Ga}$ , $^{76}\text{Ge}$ , $^{76}\text{As}$ , $^{76}\text{Se}$                    | $5.175386 \times 10^{-56}$ | D2    |
| 77 | 13 | $^{77}\text{Ti}$ , $^{77}\text{V}$ , $^{77}\text{Cr}$ , $^{77}\text{Mn}$ , $^{77}\text{Fe}$ , $^{77}\text{Co}$ , $^{77}\text{Ni}$ , $^{77}\text{Cu}$ , $^{77}\text{Zn}$ , $^{77}\text{Ga}$ , $^{77}\text{Ge}$ , $^{77}\text{As}$ , $^{77}\text{Se}$                    | $1.020116 \times 10^{-55}$ | D2    |
| 78 | 12 | $^{78}\text{V}$ , $^{78}\text{Cr}$ , $^{78}\text{Mn}$ , $^{78}\text{Fe}$ , $^{78}\text{Co}$ , $^{78}\text{Ni}$ , $^{78}\text{Cu}$ , $^{78}\text{Zn}$ , $^{78}\text{Ga}$ , $^{78}\text{Ge}$ , $^{78}\text{As}$ , $^{78}\text{Se}$                                       | $2.107071 \times 10^{-55}$ | D2    |
| 79 | 13 | $^{79}\text{V}$ , $^{79}\text{Cr}$ , $^{79}\text{Mn}$ , $^{79}\text{Fe}$ , $^{79}\text{Co}$ , $^{79}\text{Ni}$ , $^{79}\text{Cu}$ , $^{79}\text{Zn}$ , $^{79}\text{Ga}$ , $^{79}\text{Ge}$ , $^{79}\text{As}$ , $^{79}\text{Se}$ , $^{79}\text{Br}$                    | $5.182884 \times 10^{-56}$ | D2    |
| 80 | 14 | $^{80}\text{V}$ , $^{80}\text{Cr}$ , $^{80}\text{Mn}$ , $^{80}\text{Fe}$ , $^{80}\text{Co}$ , $^{80}\text{Ni}$ , $^{80}\text{Cu}$ , $^{80}\text{Zn}$ , $^{80}\text{Ga}$ , $^{80}\text{Ge}$ , $^{80}\text{As}$ , $^{80}\text{Se}$ , $^{80}\text{Br}$ , $^{80}\text{Kr}$ | $1.016253 \times 10^{-55}$ | D2    |
| 81 | 13 | $^{81}\text{Cr}$ , $^{81}\text{Mn}$ , $^{81}\text{Fe}$ , $^{81}\text{Co}$ , $^{81}\text{Ni}$ , $^{81}\text{Cu}$ , $^{81}\text{Zn}$ , $^{81}\text{Ga}$ , $^{81}\text{Ge}$ , $^{81}\text{As}$ , $^{81}\text{Se}$ , $^{81}\text{Br}$ , $^{81}\text{Kr}$                   | $1.210256 \times 10^{-55}$ | D2    |
| 82 | 13 | $^{82}\text{Cr}$ , $^{82}\text{Mn}$ , $^{82}\text{Fe}$ , $^{82}\text{Co}$ , $^{82}\text{Ni}$ , $^{82}\text{Cu}$ , $^{82}\text{Zn}$ , $^{82}\text{Ga}$ , $^{82}\text{Ge}$ , $^{82}\text{As}$ , $^{82}\text{Se}$ , $^{82}\text{Br}$ , $^{82}\text{Kr}$                   | $1.046129 \times 10^{-55}$ | D2    |
| 83 | 13 | $^{83}\text{Cr}$ , $^{83}\text{Mn}$ , $^{83}\text{Fe}$ , $^{83}\text{Co}$ , $^{83}\text{Ni}$ , $^{83}\text{Cu}$ , $^{83}\text{Zn}$ , $^{83}\text{Ga}$ , $^{83}\text{Ge}$ , $^{83}\text{As}$ , $^{83}\text{Se}$ , $^{83}\text{Br}$ , $^{83}\text{Kr}$                   | $1.597622 \times 10^{-55}$ | D2    |

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| A   | N  | Represented nuclides                                                                                                                                                                                                                                                                                                         | Archived $Y_A$             | Class |
|-----|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------|
| 84  | 13 | $^{84}\text{Cr}$ , $^{84}\text{Mn}$ , $^{84}\text{Fe}$ , $^{84}\text{Co}$ , $^{84}\text{Ni}$ , $^{84}\text{Cu}$ , $^{84}\text{Zn}$ , $^{84}\text{Ga}$ , $^{84}\text{Ge}$ , $^{84}\text{As}$ , $^{84}\text{Se}$ , $^{84}\text{Br}$ , $^{84}\text{Kr}$                                                                         | $2.743322 \times 10^{-55}$ | D2    |
| 85  | 13 | $^{85}\text{Mn}$ , $^{85}\text{Fe}$ , $^{85}\text{Co}$ , $^{85}\text{Ni}$ , $^{85}\text{Cu}$ , $^{85}\text{Zn}$ , $^{85}\text{Ga}$ , $^{85}\text{Ge}$ , $^{85}\text{As}$ , $^{85}\text{Se}$ , $^{85}\text{Br}$ , $^{85}\text{Kr}$ , $^{85}\text{Rb}$                                                                         | $2.316254 \times 10^{-55}$ | D2    |
| 86  | 14 | $^{86}\text{Mn}$ , $^{86}\text{Fe}$ , $^{86}\text{Co}$ , $^{86}\text{Ni}$ , $^{86}\text{Cu}$ , $^{86}\text{Zn}$ , $^{86}\text{Ga}$ , $^{86}\text{Ge}$ , $^{86}\text{As}$ , $^{86}\text{Se}$ , $^{86}\text{Br}$ , $^{86}\text{Kr}$ , $^{86}\text{Rb}$ , $^{86}\text{Sr}$                                                      | $1.44283 \times 10^{-55}$  | D2    |
| 87  | 14 | $^{87}\text{Mn}$ , $^{87}\text{Fe}$ , $^{87}\text{Co}$ , $^{87}\text{Ni}$ , $^{87}\text{Cu}$ , $^{87}\text{Zn}$ , $^{87}\text{Ga}$ , $^{87}\text{Ge}$ , $^{87}\text{As}$ , $^{87}\text{Se}$ , $^{87}\text{Br}$ , $^{87}\text{Kr}$ , $^{87}\text{Rb}$ , $^{87}\text{Sr}$                                                      | $6.053177 \times 10^{-56}$ | D2    |
| 88  | 13 | $^{88}\text{Fe}$ , $^{88}\text{Co}$ , $^{88}\text{Ni}$ , $^{88}\text{Cu}$ , $^{88}\text{Zn}$ , $^{88}\text{Ga}$ , $^{88}\text{Ge}$ , $^{88}\text{As}$ , $^{88}\text{Se}$ , $^{88}\text{Br}$ , $^{88}\text{Kr}$ , $^{88}\text{Rb}$ , $^{88}\text{Sr}$                                                                         | $2.926046 \times 10^{-56}$ | D2    |
| 89  | 14 | $^{89}\text{Fe}$ , $^{89}\text{Co}$ , $^{89}\text{Ni}$ , $^{89}\text{Cu}$ , $^{89}\text{Zn}$ , $^{89}\text{Ga}$ , $^{89}\text{Ge}$ , $^{89}\text{As}$ , $^{89}\text{Se}$ , $^{89}\text{Br}$ , $^{89}\text{Kr}$ , $^{89}\text{Rb}$ , $^{89}\text{Sr}$ , $^{89}\text{Y}$                                                       | $6.1468 \times 10^{-56}$   | D2    |
| 90  | 15 | $^{90}\text{Fe}$ , $^{90}\text{Co}$ , $^{90}\text{Ni}$ , $^{90}\text{Cu}$ , $^{90}\text{Zn}$ , $^{90}\text{Ga}$ , $^{90}\text{Ge}$ , $^{90}\text{As}$ , $^{90}\text{Se}$ , $^{90}\text{Br}$ , $^{90}\text{Kr}$ , $^{90}\text{Rb}$ , $^{90}\text{Sr}$ , $^{90}\text{Y}$ , $^{90}\text{Zr}$                                    | $4.745019 \times 10^{-56}$ | D2    |
| 91  | 14 | $^{91}\text{Co}$ , $^{91}\text{Ni}$ , $^{91}\text{Cu}$ , $^{91}\text{Zn}$ , $^{91}\text{Ga}$ , $^{91}\text{Ge}$ , $^{91}\text{As}$ , $^{91}\text{Se}$ , $^{91}\text{Br}$ , $^{91}\text{Kr}$ , $^{91}\text{Rb}$ , $^{91}\text{Sr}$ , $^{91}\text{Y}$ , $^{91}\text{Zr}$                                                       | $4.245786 \times 10^{-56}$ | D2    |
| 92  | 14 | $^{92}\text{Co}$ , $^{92}\text{Ni}$ , $^{92}\text{Cu}$ , $^{92}\text{Zn}$ , $^{92}\text{Ga}$ , $^{92}\text{Ge}$ , $^{92}\text{As}$ , $^{92}\text{Se}$ , $^{92}\text{Br}$ , $^{92}\text{Kr}$ , $^{92}\text{Rb}$ , $^{92}\text{Sr}$ , $^{92}\text{Y}$ , $^{92}\text{Zr}$                                                       | $3.309035 \times 10^{-56}$ | D2    |
| 93  | 15 | $^{93}\text{Co}$ , $^{93}\text{Ni}$ , $^{93}\text{Cu}$ , $^{93}\text{Zn}$ , $^{93}\text{Ga}$ , $^{93}\text{Ge}$ , $^{93}\text{As}$ , $^{93}\text{Se}$ , $^{93}\text{Br}$ , $^{93}\text{Kr}$ , $^{93}\text{Rb}$ , $^{93}\text{Sr}$ , $^{93}\text{Y}$ , $^{93}\text{Zr}$ , $^{93}\text{Nb}$                                    | $2.900263 \times 10^{-56}$ | D2    |
| 94  | 15 | $^{94}\text{Ni}$ , $^{94}\text{Cu}$ , $^{94}\text{Zn}$ , $^{94}\text{Ga}$ , $^{94}\text{Ge}$ , $^{94}\text{As}$ , $^{94}\text{Se}$ , $^{94}\text{Br}$ , $^{94}\text{Kr}$ , $^{94}\text{Rb}$ , $^{94}\text{Sr}$ , $^{94}\text{Y}$ , $^{94}\text{Zr}$ , $^{94}\text{Nb}$ , $^{94}\text{Mo}$                                    | $3.931671 \times 10^{-56}$ | D2    |
| 95  | 15 | $^{95}\text{Ni}$ , $^{95}\text{Cu}$ , $^{95}\text{Zn}$ , $^{95}\text{Ga}$ , $^{95}\text{Ge}$ , $^{95}\text{As}$ , $^{95}\text{Se}$ , $^{95}\text{Br}$ , $^{95}\text{Kr}$ , $^{95}\text{Rb}$ , $^{95}\text{Sr}$ , $^{95}\text{Y}$ , $^{95}\text{Zr}$ , $^{95}\text{Nb}$ , $^{95}\text{Mo}$                                    | $2.978948 \times 10^{-56}$ | D2    |
| 96  | 15 | $^{96}\text{Ni}$ , $^{96}\text{Cu}$ , $^{96}\text{Zn}$ , $^{96}\text{Ga}$ , $^{96}\text{Ge}$ , $^{96}\text{As}$ , $^{96}\text{Se}$ , $^{96}\text{Br}$ , $^{96}\text{Kr}$ , $^{96}\text{Rb}$ , $^{96}\text{Sr}$ , $^{96}\text{Y}$ , $^{96}\text{Zr}$ , $^{96}\text{Nb}$ , $^{96}\text{Mo}$                                    | $1.634314 \times 10^{-56}$ | D2    |
| 97  | 14 | $^{97}\text{Cu}$ , $^{97}\text{Zn}$ , $^{97}\text{Ga}$ , $^{97}\text{Ge}$ , $^{97}\text{As}$ , $^{97}\text{Se}$ , $^{97}\text{Br}$ , $^{97}\text{Kr}$ , $^{97}\text{Rb}$ , $^{97}\text{Sr}$ , $^{97}\text{Y}$ , $^{97}\text{Zr}$ , $^{97}\text{Nb}$ , $^{97}\text{Mo}$                                                       | $2.386422 \times 10^{-56}$ | D2    |
| 98  | 14 | $^{98}\text{Cu}$ , $^{98}\text{Zn}$ , $^{98}\text{Ga}$ , $^{98}\text{Ge}$ , $^{98}\text{As}$ , $^{98}\text{Se}$ , $^{98}\text{Br}$ , $^{98}\text{Kr}$ , $^{98}\text{Rb}$ , $^{98}\text{Sr}$ , $^{98}\text{Y}$ , $^{98}\text{Zr}$ , $^{98}\text{Nb}$ , $^{98}\text{Mo}$                                                       | $1.19575 \times 10^{-56}$  | D2    |
| 99  | 16 | $^{99}\text{Cu}$ , $^{99}\text{Zn}$ , $^{99}\text{Ga}$ , $^{99}\text{Ge}$ , $^{99}\text{As}$ , $^{99}\text{Se}$ , $^{99}\text{Br}$ , $^{99}\text{Kr}$ , $^{99}\text{Rb}$ , $^{99}\text{Sr}$ , $^{99}\text{Y}$ , $^{99}\text{Zr}$ , $^{99}\text{Nb}$ , $^{99}\text{Mo}$ , $^{99}\text{Tc}$ , $^{99}\text{Ru}$                 | $4.010949 \times 10^{-57}$ | D2    |
| 100 | 16 | $^{100}\text{Cu}$ , $^{100}\text{Zn}$ , $^{100}\text{Ga}$ , $^{100}\text{Ge}$ , $^{100}\text{As}$ , $^{100}\text{Se}$ , $^{100}\text{Br}$ , $^{100}\text{Kr}$ , $^{100}\text{Rb}$ , $^{100}\text{Sr}$ , $^{100}\text{Y}$ , $^{100}\text{Zr}$ , $^{100}\text{Nb}$ , $^{100}\text{Mo}$ , $^{100}\text{Tc}$ , $^{100}\text{Ru}$ | $4.66066 \times 10^{-57}$  | D2    |
| 101 | 15 | $^{101}\text{Zn}$ , $^{101}\text{Ga}$ , $^{101}\text{Ge}$ , $^{101}\text{As}$ , $^{101}\text{Se}$ , $^{101}\text{Br}$ , $^{101}\text{Kr}$ , $^{101}\text{Rb}$ , $^{101}\text{Sr}$ , $^{101}\text{Y}$ , $^{101}\text{Zr}$ , $^{101}\text{Nb}$ , $^{101}\text{Mo}$ , $^{101}\text{Tc}$ , $^{101}\text{Ru}$                     | $3.713053 \times 10^{-57}$ | D2    |

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| A   | N  | Represented nuclides                                                                                                                                                                                                                                                                                                                                | Archived $Y_A$             | Class |
|-----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------|
| 102 | 15 | $^{102}\text{Zn}$ , $^{102}\text{Ga}$ , $^{102}\text{Ge}$ , $^{102}\text{As}$ , $^{102}\text{Se}$ , $^{102}\text{Br}$ , $^{102}\text{Kr}$ , $^{102}\text{Rb}$ , $^{102}\text{Sr}$ , $^{102}\text{Y}$ , $^{102}\text{Zr}$ ,<br>$^{102}\text{Nb}$ , $^{102}\text{Mo}$ , $^{102}\text{Tc}$ , $^{102}\text{Ru}$                                         | $8.071407 \times 10^{-57}$ | D2    |
| 103 | 16 | $^{103}\text{Zn}$ , $^{103}\text{Ga}$ , $^{103}\text{Ge}$ , $^{103}\text{As}$ , $^{103}\text{Se}$ , $^{103}\text{Br}$ , $^{103}\text{Kr}$ , $^{103}\text{Rb}$ , $^{103}\text{Sr}$ , $^{103}\text{Y}$ , $^{103}\text{Zr}$ ,<br>$^{103}\text{Nb}$ , $^{103}\text{Mo}$ , $^{103}\text{Tc}$ , $^{103}\text{Ru}$ , $^{103}\text{Rh}$                     | $2.487525 \times 10^{-57}$ | D2    |
| 104 | 16 | $^{104}\text{Ga}$ , $^{104}\text{Ge}$ , $^{104}\text{As}$ , $^{104}\text{Se}$ , $^{104}\text{Br}$ , $^{104}\text{Kr}$ , $^{104}\text{Rb}$ , $^{104}\text{Sr}$ , $^{104}\text{Y}$ , $^{104}\text{Zr}$ , $^{104}\text{Nb}$ ,<br>$^{104}\text{Mo}$ , $^{104}\text{Tc}$ , $^{104}\text{Ru}$ , $^{104}\text{Rh}$ , $^{104}\text{Pd}$                     | $6.492919 \times 10^{-57}$ | D2    |
| 105 | 16 | $^{105}\text{Ga}$ , $^{105}\text{Ge}$ , $^{105}\text{As}$ , $^{105}\text{Se}$ , $^{105}\text{Br}$ , $^{105}\text{Kr}$ , $^{105}\text{Rb}$ , $^{105}\text{Sr}$ , $^{105}\text{Y}$ , $^{105}\text{Zr}$ , $^{105}\text{Nb}$ ,<br>$^{105}\text{Mo}$ , $^{105}\text{Tc}$ , $^{105}\text{Ru}$ , $^{105}\text{Rh}$ , $^{105}\text{Pd}$                     | $8.021328 \times 10^{-57}$ | D2    |
| 106 | 16 | $^{106}\text{Ga}$ , $^{106}\text{Ge}$ , $^{106}\text{As}$ , $^{106}\text{Se}$ , $^{106}\text{Br}$ , $^{106}\text{Kr}$ , $^{106}\text{Rb}$ , $^{106}\text{Sr}$ , $^{106}\text{Y}$ , $^{106}\text{Zr}$ , $^{106}\text{Nb}$ ,<br>$^{106}\text{Mo}$ , $^{106}\text{Tc}$ , $^{106}\text{Ru}$ , $^{106}\text{Rh}$ , $^{106}\text{Pd}$                     | $3.052984 \times 10^{-57}$ | D2    |
| 107 | 16 | $^{107}\text{Ge}$ , $^{107}\text{As}$ , $^{107}\text{Se}$ , $^{107}\text{Br}$ , $^{107}\text{Kr}$ , $^{107}\text{Rb}$ , $^{107}\text{Sr}$ , $^{107}\text{Y}$ , $^{107}\text{Zr}$ , $^{107}\text{Nb}$ , $^{107}\text{Mo}$ ,<br>$^{107}\text{Tc}$ , $^{107}\text{Ru}$ , $^{107}\text{Rh}$ , $^{107}\text{Pd}$ , $^{107}\text{Ag}$                     | $4.27056 \times 10^{-57}$  | D2    |
| 108 | 17 | $^{108}\text{Ge}$ , $^{108}\text{As}$ , $^{108}\text{Se}$ , $^{108}\text{Br}$ , $^{108}\text{Kr}$ , $^{108}\text{Rb}$ , $^{108}\text{Sr}$ , $^{108}\text{Y}$ , $^{108}\text{Zr}$ , $^{108}\text{Nb}$ , $^{108}\text{Mo}$ ,<br>$^{108}\text{Tc}$ , $^{108}\text{Ru}$ , $^{108}\text{Rh}$ , $^{108}\text{Pd}$ , $^{108}\text{Ag}$ , $^{108}\text{Cd}$ | $6.845493 \times 10^{-57}$ | D2    |
| 109 | 17 | $^{109}\text{Ge}$ , $^{109}\text{As}$ , $^{109}\text{Se}$ , $^{109}\text{Br}$ , $^{109}\text{Kr}$ , $^{109}\text{Rb}$ , $^{109}\text{Sr}$ , $^{109}\text{Y}$ , $^{109}\text{Zr}$ , $^{109}\text{Nb}$ , $^{109}\text{Mo}$ ,<br>$^{109}\text{Tc}$ , $^{109}\text{Ru}$ , $^{109}\text{Rh}$ , $^{109}\text{Pd}$ , $^{109}\text{Ag}$ , $^{109}\text{Cd}$ | $2.022383 \times 10^{-57}$ | D2    |
| 110 | 16 | $^{110}\text{As}$ , $^{110}\text{Se}$ , $^{110}\text{Br}$ , $^{110}\text{Kr}$ , $^{110}\text{Rb}$ , $^{110}\text{Sr}$ , $^{110}\text{Y}$ , $^{110}\text{Zr}$ , $^{110}\text{Nb}$ , $^{110}\text{Mo}$ , $^{110}\text{Tc}$ ,<br>$^{110}\text{Ru}$ , $^{110}\text{Rh}$ , $^{110}\text{Pd}$ , $^{110}\text{Ag}$ , $^{110}\text{Cd}$                     | $5.998031 \times 10^{-57}$ | D2    |
| 111 | 16 | $^{111}\text{As}$ , $^{111}\text{Se}$ , $^{111}\text{Br}$ , $^{111}\text{Kr}$ , $^{111}\text{Rb}$ , $^{111}\text{Sr}$ , $^{111}\text{Y}$ , $^{111}\text{Zr}$ , $^{111}\text{Nb}$ , $^{111}\text{Mo}$ , $^{111}\text{Tc}$ ,<br>$^{111}\text{Ru}$ , $^{111}\text{Rh}$ , $^{111}\text{Pd}$ , $^{111}\text{Ag}$ , $^{111}\text{Cd}$                     | $2.259657 \times 10^{-57}$ | D2    |
| 112 | 16 | $^{112}\text{As}$ , $^{112}\text{Se}$ , $^{112}\text{Br}$ , $^{112}\text{Kr}$ , $^{112}\text{Rb}$ , $^{112}\text{Sr}$ , $^{112}\text{Y}$ , $^{112}\text{Zr}$ , $^{112}\text{Nb}$ , $^{112}\text{Mo}$ , $^{112}\text{Tc}$ ,<br>$^{112}\text{Ru}$ , $^{112}\text{Rh}$ , $^{112}\text{Pd}$ , $^{112}\text{Ag}$ , $^{112}\text{Cd}$                     | $6.772206 \times 10^{-57}$ | D2    |
| 113 | 16 | $^{113}\text{Se}$ , $^{113}\text{Br}$ , $^{113}\text{Kr}$ , $^{113}\text{Rb}$ , $^{113}\text{Sr}$ , $^{113}\text{Y}$ , $^{113}\text{Zr}$ , $^{113}\text{Nb}$ , $^{113}\text{Mo}$ , $^{113}\text{Tc}$ , $^{113}\text{Ru}$ ,<br>$^{113}\text{Rh}$ , $^{113}\text{Pd}$ , $^{113}\text{Ag}$ , $^{113}\text{Cd}$ , $^{113}\text{In}$                     | $1.479664 \times 10^{-57}$ | D2    |
| 114 | 17 | $^{114}\text{Se}$ , $^{114}\text{Br}$ , $^{114}\text{Kr}$ , $^{114}\text{Rb}$ , $^{114}\text{Sr}$ , $^{114}\text{Y}$ , $^{114}\text{Zr}$ , $^{114}\text{Nb}$ , $^{114}\text{Mo}$ , $^{114}\text{Tc}$ , $^{114}\text{Ru}$ ,<br>$^{114}\text{Rh}$ , $^{114}\text{Pd}$ , $^{114}\text{Ag}$ , $^{114}\text{Cd}$ , $^{114}\text{In}$ , $^{114}\text{Sn}$ | $7.965147 \times 10^{-57}$ | D2    |
| 115 | 17 | $^{115}\text{Se}$ , $^{115}\text{Br}$ , $^{115}\text{Kr}$ , $^{115}\text{Rb}$ , $^{115}\text{Sr}$ , $^{115}\text{Y}$ , $^{115}\text{Zr}$ , $^{115}\text{Nb}$ , $^{115}\text{Mo}$ , $^{115}\text{Tc}$ , $^{115}\text{Ru}$ ,<br>$^{115}\text{Rh}$ , $^{115}\text{Pd}$ , $^{115}\text{Ag}$ , $^{115}\text{Cd}$ , $^{115}\text{In}$ , $^{115}\text{Sn}$ | $3.090037 \times 10^{-57}$ | D2    |
| 116 | 17 | $^{116}\text{Se}$ , $^{116}\text{Br}$ , $^{116}\text{Kr}$ , $^{116}\text{Rb}$ , $^{116}\text{Sr}$ , $^{116}\text{Y}$ , $^{116}\text{Zr}$ , $^{116}\text{Nb}$ , $^{116}\text{Mo}$ , $^{116}\text{Tc}$ , $^{116}\text{Ru}$ ,<br>$^{116}\text{Rh}$ , $^{116}\text{Pd}$ , $^{116}\text{Ag}$ , $^{116}\text{Cd}$ , $^{116}\text{In}$ , $^{116}\text{Sn}$ | $2.0995 \times 10^{-57}$   | D2    |
| 117 | 16 | $^{117}\text{Br}$ , $^{117}\text{Kr}$ , $^{117}\text{Rb}$ , $^{117}\text{Sr}$ , $^{117}\text{Y}$ , $^{117}\text{Zr}$ , $^{117}\text{Nb}$ , $^{117}\text{Mo}$ , $^{117}\text{Tc}$ , $^{117}\text{Ru}$ , $^{117}\text{Rh}$ ,<br>$^{117}\text{Pd}$ , $^{117}\text{Ag}$ , $^{117}\text{Cd}$ , $^{117}\text{In}$ , $^{117}\text{Sn}$                     | $5.006565 \times 10^{-57}$ | D2    |
| 118 | 16 | $^{118}\text{Br}$ , $^{118}\text{Kr}$ , $^{118}\text{Rb}$ , $^{118}\text{Sr}$ , $^{118}\text{Y}$ , $^{118}\text{Zr}$ , $^{118}\text{Nb}$ , $^{118}\text{Mo}$ , $^{118}\text{Tc}$ , $^{118}\text{Ru}$ , $^{118}\text{Rh}$ ,<br>$^{118}\text{Pd}$ , $^{118}\text{Ag}$ , $^{118}\text{Cd}$ , $^{118}\text{In}$ , $^{118}\text{Sn}$                     | $7.760345 \times 10^{-57}$ | D2    |
| 119 | 16 | $^{119}\text{Br}$ , $^{119}\text{Kr}$ , $^{119}\text{Rb}$ , $^{119}\text{Sr}$ , $^{119}\text{Y}$ , $^{119}\text{Zr}$ , $^{119}\text{Nb}$ , $^{119}\text{Mo}$ , $^{119}\text{Tc}$ , $^{119}\text{Ru}$ , $^{119}\text{Rh}$ ,<br>$^{119}\text{Pd}$ , $^{119}\text{Ag}$ , $^{119}\text{Cd}$ , $^{119}\text{In}$ , $^{119}\text{Sn}$                     | $2.873667 \times 10^{-57}$ | D2    |

## 5 The Mass-Five Gap

Mass five is present in the continuous register so that the data contain no missing mass row. The two mass-five resonances explicitly represented in this construction,

$${}^5\text{He} \quad \text{and} \quad {}^5\text{Li}, \quad (11)$$

are unbound resonances:

$${}^5\text{He} \longrightarrow {}^4\text{He} + n, \quad (12)$$

$${}^5\text{Li} \longrightarrow {}^4\text{He} + p. \quad (13)$$

They are broad, particle-unstable resonances and do not accumulate as persistent abundance-bearing states under the conditions represented here. The row is not an exhaustive catalog of every evaluated  $A = 5$  system. It records the two resonances relevant to the represented  $n + {}^4\text{He}$  and  $p + {}^4\text{He}$  channels and assigns neither an endpoint abundance nor an independent native coordinate. The network can bypass the gap through channels such as

$${}^2\text{H} + {}^4\text{He} \longrightarrow {}^6\text{Li} + \gamma, \quad (14)$$

$${}^3\text{H} + {}^4\text{He} \longrightarrow {}^7\text{Li} + \gamma, \quad (15)$$

$${}^3\text{He} + {}^4\text{He} \longrightarrow {}^7\text{Be} + \gamma. \quad (16)$$

The mass-five row is consequently a physical gap designation, not a missing-data entry.

## 6 Completeness and Support Classification

The continuous register satisfies the following checks:

$$N_{\text{mass rows}} = 119, \quad (17)$$

$$N_{\text{missing mass rows}} = 0, \quad (18)$$

$$N_{\text{listed nuclide/resonance entries}} = 1111, \quad (19)$$

$$N_{\text{zero endpoint graph nodes}} = 3, \quad (20)$$

$$A_{\text{min}} = 1, \quad (21)$$

$$A_{\text{max}} = 119. \quad (22)$$

The evidence classification is

$$A = 1\text{--}30 : \text{native inventory}, \quad (23)$$

$$A = 31\text{--}32 : \text{strong prescribed-trajectory diagnostic}, \quad (24)$$

$$A = 33\text{--}119 : \text{exploratory prescribed-trajectory graph}. \quad (25)$$

The nuclide lists at masses 31 through 119 describe membership in the reaction graph. They are not claims that all listed nuclides obtain measurable primordial abundances, nor are they a catalog of every experimentally known nuclide at those masses. Some ReacLib graph nodes are supported by evaluated or theoretical rates; graph membership alone is not an experimental-discovery classification. The companion nuclide-level CSV preserves the same register in machine-readable form, including  $A$ , nuclide name, proton number, endpoint abundance when applicable, evidence class, and the explicit mass-five resonance note. The three zero-endpoint graph nodes are retained because they are members of the generated reaction graph even though their final floating-point abundance is zero.

## 7 Scientific Boundary

The complete register demonstrates continuity of the represented calculation but does not erase the distinction between native-network and diagnostic support. Native-network coverage presently ends at silicon-30. The first post-silicon block is temporally well controlled on the prescribed trajectory. The deeper tail remains an exploratory reduced-graph result.

The boundary tests show that moving the absorbing boundary from  $A = 120$  to  $A = 140$  and  $A = 160$  leaves all 89 mass-sector sums from  $A = 31$  through  $A = 119$  exactly unchanged. The qualification also establishes that downstream states begin at exact zero and vanish identically when the  $^{30}\text{Si}$  source is removed. However, source-cutoff tests show that the deep-tail magnitudes depend strongly on the early sub- $10^{-50}$  portion of that native source trajectory. This dependence does not erase graph reachability, but it prevents the smallest tabulated values from being interpreted as precision-qualified physical abundances.

Accordingly, the current result is:

The combined native and prescribed-trajectory calculation contains an explicit, gap-free archived mass register from  $A = 1$  through  $A = 119$ . The separate  $A_{\text{max}} = 160$  boundary-and-convergence qualification leaves the published  $A = 31$ – $119$  results exactly unchanged and retains positive diagnostic support through its final interior mass,  $A = 159$ . The  $A = 120$ – $159$  qualification sectors are not presented as an extension of the published register. The result is a reduced-graph reachability test; it does not identify a finite physical endpoint or establish measurable primordial heavy-element yields.

## 8 Conclusion

This addendum preserves the complete native and diagnostic mass register, documents the numerical basis of the post-silicon continuation, and adds source-isolation, source-cutoff, temporal-convergence, and outward-boundary tests.

Its principal conclusion is that the tested native frontier at silicon-30 is not a topological endpoint of the selected reduced graph. The archived calculation contains a gap-free mass-sector register from  $A = 1$  through  $A = 119$ , and the original  $A = 31$ – $119$  results are exactly unchanged when the absorbing boundary is moved from  $A = 120$  to  $A = 140$  and  $A = 160$ . Positive support also persists at 64-substep resolution and after removing early  $^{30}\text{Si}$  source values below  $10^{-50}$ , although the magnitude of the deep tail is highly source-sensitive.

The enlarged  $A_{\text{max}} = 160$  calculation is a boundary-and-convergence qualification, not a published register extension through  $A = 159$ . The present result therefore establishes robust post-silicon reachability within the selected one-way neutron-capture and beta-minus graph, but it does not establish precision-qualified heavy-element yields, a finite physical endpoint, or native-network coverage beyond silicon-30.

Because the seed calculation inherits parameters introduced previously in the CPTG gravitational comparison framework, the result also provides supporting evidence for cross-scale structural continuity. The full interpretation and validation of that connection belong to the companion universal-theory paper rather than to this exploratory extension addendum.

## 9 Data and Code Availability

The controlling release bundle is:

CPTG\_Post016\_Si30Bridge\_AbsoluteLimitDiagnostic\_20260716\_r01.zip

with SHA-256:

76ee1c5c637069972c231cfda721afbcae1fb8975ca6a128ac2bea5abf79410a

The extended qualification bundle is:

CPTG\_Post016\_Si30FloorBoundaryConvergenceQualification\_20260719\_r01.zip

with SHA-256:

908642caa9dbac8557faba937abcb9d1e488e3cf1d7debb787db27aa3df2c60e

Within the CPTG repository, the repository-relative release location is:

<https://github.com/CLG2025/CPTG/tree/main/nuclear-reactions>

The principal machine-readable inputs and implementation files are:

- RESULTS/TRACE\_DIAGNOSTIC/raw\_endpoint.csv  
SHA-256 9e234e2d8f672ad77fbfa1faa26c7fe20020e62325f07eabaa2ae5fdae1272e7
- RESULTS/TRACE\_DIAGNOSTIC/SI30\_NEUTRON\_BETA\_A120\_S8\_TRI\_ENDPOINT.csv  
SHA-256 aecc87d9a58b5e16f2977c2857ddcd4045fd56cf5172ea397b90d1808cda8180
- RESULTS/TRACE\_DIAGNOSTIC/SI30\_NEUTRON\_BETA\_A120\_S8\_TRI\_BY\_MASS.csv  
SHA-256 a96b8914c977766e981e017a0054c40b52935a6fa5ab1a2aaf04a3567a797817
- SOURCE/si30\_neutron\_beta\_triangular\_fast.py  
SHA-256 9adb230db299020e7be09fc4e5308a8af70814939242fd9b762e50ec0f73771c
- SOURCE/si30\_floor\_boundary\_convergence\_qualification.py  
SHA-256 99cb37898e65698c482c90feaea64cfa3928c95fa1a6cafb9b40b96ee694c89b
- RESULTS/QUALIFICATION\_SUMMARY.json  
source, boundary, convergence, and cutoff decision record

The upstream release identities retained in the bundle are:

- CPTG\_Post016\_TheoryExtension\_AI\_Handoff\_20260716\_r01.zip  
SHA-256 20cbaca5676f9d06dd1cea2be6b45c9c1ecd60dfc07b75032cdc1221b2330c4a
- CPTG\_Post016\_RobustFrontierConfirmation\_20260716\_r02.zip  
SHA-256 75a1855ebf7e3460095ace4383d9c704d161a5f5a7a01cc75f1184f6a3688662
- CPTG\_Ne23\_BBN\_vs\_ExplosiveRegime\_Discrimination\_20260716\_r01.zip  
SHA-256 8757fa29971af6f6375978ee6731b6d04c425e0916bfafc5c6a4de512ef10b60

The native trajectory was produced with AlterBBN v2.2 at  $\eta_{10} = 6.122$ , `failsafe=6`, and abundance floor  $10^{-50}$ . The archived prescribed-trajectory continuation used Python 3.13.5, `pynucastro` 2.11.0, a maximum network mass of  $A = 120$ , and background-interpolated recursive backward Euler. The qualification used the same Python and `pynucastro` versions, extended the graph to  $A = 160$ , and applied the identical recurrence with Numba 0.65.1 acceleration. The qualification source contains the public nuclear-data inputs, stored native trajectories, and numerical recurrence required to reproduce

the reported output-layer tests.

## References

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